AGENTIC HACKATHON – ALPACA PAPER TRADING

Momentum + Options Agents

Five autonomous agents. One risk officer with veto power.

THE PITCH

Discipline doesn't scale by hand

Momentum investing works – but running it with real discipline means position sizing, earnings-drift detection, options-overlay collateral math, risk vetoes, and exit timing, all done consistently, every single day.

Most people can't hold six agents' worth of judgment in their head at once. This system runs it as a pipeline of five specialized agents (plus an optional sixth) – each with one narrow job, so no single component can do everything, and nothing gets forgotten.

ARCHITECTURE

The six-agent pipeline

1. ***Market Data Agent***
   Scores momentum, reads real SEC earnings filings
2. ***Strategy Agent***
   Proposes buys / trims / exits + option spreads (rules only)
3. OPTIONAL
   Judgment Agent (AI)
   Can flag a name as risky. Cannot invent or size a trade. Optional – abstains w/o API key
4. ***Risk Agent***
   VETO AUTHORITY.
   Position limits, per-name caps, single-pot capital rule
5. ***Execution Agent***
   ONLY agent that talks to the broker. Refuses unapproved orders – raises, not submits
6. ***Position Agent***
   Reports P&L, exposure, reconciles the book

"The one-way valve: only Execution can write to the broker, and it structurally cannot bypass Risk."

THE STRATEGY

Three sleeves, one pot of capital

MOMENTUM CORE

80%

Top 6 stocks by vol-adjusted momentum score (12m minus last month, 3-month, trend, relative strength vs SPY). Equal-weighted, rebalanced periodically. Goes to 100% cash if SPY is below its 200-day moving average.

PEAD SLEEVE

5% / event

Buys stocks the day after a confirmed earnings beat, using REAL SEC EDGAR filing data (not simulated) – measured two independent ways (SUE score + volume-confirmed price gap) that both must agree.

OPTIONS OVERLAY

15% max collateral

Sells 20-delta put credit spreads on stocks the other sleeves already own – collects premium on names it's already bullish on. Closes each spread at 50% of max profit, or unconditionally within 5 days of expiry to avoid assignment.

BACKTEST RESULTS

Train / validate / test — never re-tuned

Period	Return	Win Rate	Profit Factor	Max Drawdown	Ann. Sharpe	Alpha (ann)	vs SPY
Train 2011-2018	+360.1%	95.2%	2.12	-15.3%	0.95	+12.0%	+131.3%
Validate 2019-2021	+221.7%	92.7%	1.20	-47.0%	1.17	+32.0%	+99.7%
Test 2022-2026	+1072.1%	94.3%	2.47	-30.6%	1.88	+64.5%	+63.7%

Test period was run once and never tuned on.

INTELLECTUAL HONESTY

How much of that return is universe selection?

On a mechanically-screened 951-stock universe (survivorship-bias-free, includes delisted/bankrupt names) vs. the hand-picked 39-name universe behind the headline numbers, 2021–2026:

SPY (baseline)	951-name universe	Hand-picked 39 names
+70.7%	**+329.2%**	**+249.1%**
Sharpe 0.60	Sharpe 0.93 · alpha +28.7%/yr	Basis for Slide 5 headline

~40% of the headline backtest return was universe selection, not the strategy.
Expect ~+29%/yr alpha going forward – not the hand-picked number.

LIVE PROOF – THIS ACTUALLY HAPPENED

From broken fills to a real credit spread, in 6 days

Aug 28 — **Momentum core deployed live on Alpaca paper trading – bought 6 stocks (CNC, JAZZ, TD, UTHR, VLO, WBD) equal-weight. Clean fill.**

Aug 28 — **3 option spread orders (WBD, JAZZ, TD) submitted – ALL THREE expired unfilled. Root cause: limit price calculated from a Black-Scholes model's own strikes/expiry, not Alpaca's real listed contracts – every order asked for more credit than the spread could pay.**

Sep 2-3 — **Diagnosed & fixed 4 independent defects (mispriced limit vs. real contracts; overlay could fire only once per rebalance; expiry search window too narrow; duplicate-order check broken for multi-leg orders). Also found: agents could OPEN a spread but had no path to CLOSE one. Built a leg-pairing module that reconstructs an open spread from the broker's own position data.**

Sep 3, 10:33am ET — **THE TD 115/105 PUT CREDIT SPREAD FILLED – sold 115 put @ $0.95, bought 105 put @ $0.40, net credit $0.55/share ($110 total) on 2 contracts. Real premium collected.**

Sep 3, later — **12:45pm & 3:45pm scheduled cycles both correctly recognized the open position, did NOT duplicate it, and correctly left it alone – the full open hold close lifecycle running unattended on real fills, not a backtest.**

"This is the exact failure the system is fixed against, working live, autonomously, without a human in the loop."

SAFETY DESIGN

Four layers between a proposal and a fill

Paper trading only, structurally

4 independent gates must ALL pass before any order reaches the broker: env flag, paper API endpoint, account number prefix, and a SHA-256 hash of the account number checked against a tracked allowlist in source code – not just a .env file, so widening access requires a reviewable code change.

Capital cannot be double-spent

Enforced by runtime assertions, not just logic that could silently drift. An earlier bug where this wasn't enforced inflated a 15-year backtest CAGR by ~8 percentage points before being caught and fixed.

64 automated tests

Including tests that pin the capital invariants, and tests that specifically prove the Execution Agent raises an exception (refuses) if handed an unapproved order.

Silent-safe on a schedule

Runs 3x per weekday with zero human intervention required – but any run that finds nothing to do is safe by default: if the market is closed, every proposal is automatically rejected.

WHAT'S NEXT

Extending what already works

- Extend the PEAD sleeve's live coverage beyond the current names.
- Tighten the options overlay's fill rate further.
- Consider adding more sleeves to the pipeline.

Five agents. One veto. Real fills.

Momentum + Options Agents – Agentic Hackathon, Alpaca Paper Trading